1° Parcial

Alumna = Paulina Contreras   Legajo = 168642-2

1) Teoría Sist de coor Intrínseco componentes de la aceleración. Escribir vect la aceleración en func de estas componentes. Direc de valores.

~~$\vec{a} = a_t + a_n$~~

El vector aceleración se obtiene de derivar la velocidad (que tiene direcc tangencial en este sistema).

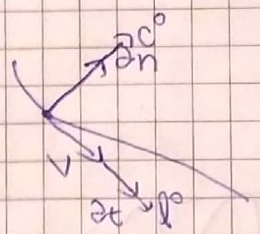
$\vec{V}(t) = V\ell^\circ$ ℓ° (versor de la tangencial)

$$\vec{a}_{(t)} = \frac{d\vec{V}(t)}{dt} = \frac{d}{dt}(V\ell^\circ) = \frac{dV}{dt}\ell^\circ + V\frac{d\ell^\circ}{dt}$$

$$a_n = \frac{V^2}{R_c} \rightarrow \text{componente normal}$$

$$a_t = \frac{dV}{dt} \rightarrow \text{componente tangencial}$$

$$\vec{a}(t) = \frac{dV}{dt} + \frac{V^2}{R_c}$$



→ La aceleración normal siempre apunta al centro de curvatura

→ La a_t puede ser positiva o neg pero acompaña siempre direc tangencial (de la velocidad.

b) $R = 3t^2\,\check{i} + 10t\,\check{j}$. Determinar componente tg de aceleración y normal para $t = 2s$

$\vec{a} = \frac{dV}{dt}$ y $V = \frac{dr}{dt} = 6t\,\check{i} + 10$

$\boxed{a = 6\,\check{i}} \rightarrow$ aceleración total

~~$a_t = \frac{d|\vec{V}|}{dt}$~~

$\vec{a} = \vec{a}_n + \vec{a}_{(t)}$

$$a_t = \frac{d|\vec{V}|}{dt} \qquad |\vec{V}| = \sqrt{(6t)^2 + 100}$$

$$|\vec{V}| = \sqrt{36t^2 + 100}$$

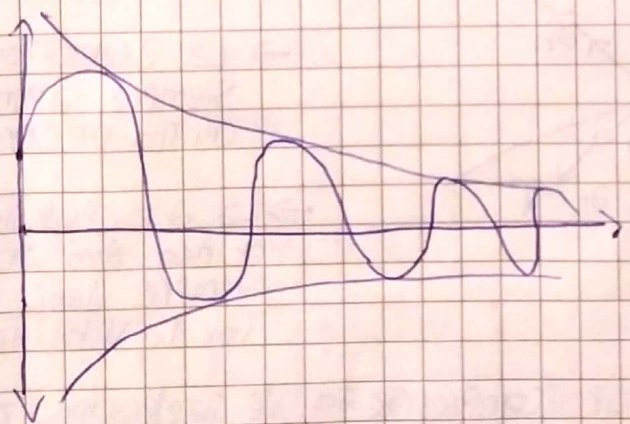
$$a_t = \frac{72t}{2\sqrt{36t^2+100}}$$

$$a_t = \frac{36t}{\sqrt{36t^2+100}} \rightarrow a_t(2) = \frac{36 \cdot 2}{2\sqrt{61}} = \boxed{4,61}$$

$$a_n = \sqrt{6^2 - 4,61^2}$$

$$\boxed{a_n = 3,84}$$

2) Relacione el amortiguamiento utilizado con el crítico (Depende del sist y de su frec natural) → es la relac entre el amortiguamiento real y el crítico $(\eta = \beta / \beta_c)$ → a mayor Amort menor Amplitud.

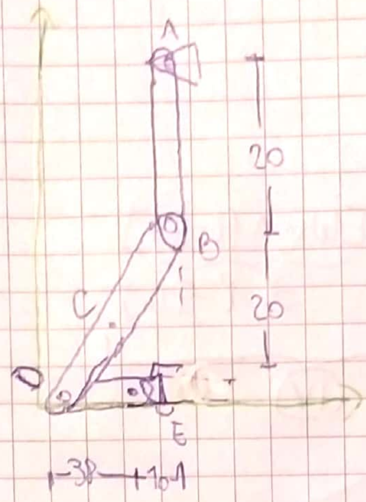


Amplitud Decreciente con el tiempo

$\rightarrow w_c > w$

Barra $\omega_{AB} = cte = -19\,s^{-1}$. A y E pivos. Determinar γ_{BD} γ_{DE}.

Dist cm.



$V_B = V_A + \omega_{AB} \wedge (B-A)$

$$V_B = \begin{vmatrix} \check{i} & \check{j} & \check{k} \\ 0 & 0 & -19 \\ 0 & -20 & 0 \end{vmatrix} = -380\,\check{i}$$

$V_D = V_B + \omega_{BD} \wedge (D-B)$

$$\begin{vmatrix} \check{i} & \check{j} & \check{k} \\ 0 & 0 & \omega_{BD} \\ -48 & -20 & 0 \end{vmatrix} = \underbrace{20\omega_{BD}\,\check{i} - 48\omega_{BD}\,\check{j}}_{380\check{i} - 912\check{j}}$$

$$\boxed{V_D = \check{i}(-380 + 20\omega_{BD}) - 48\omega_{BD}\,\check{j}} \rightarrow \text{I}$$

$V_D = \overset{0}{V_E} + \omega_{DE} \wedge (D-E)$

$$V_D = \begin{vmatrix} \check{i} & \check{j} & \check{k} \\ 0 & 0 & \omega_{DE} \\ -38 & 0 & 0 \end{vmatrix} = \boxed{-38\,\omega_{DE}\,\check{j}} \rightarrow \text{II}$$

$\rightarrow -912\,\check{j}$

I = II

$\check{i})$ $-380 + 20\omega_{BD} = 0$

$\omega_{BD} = 19\,s^{-1}\,\check{k}$

$\check{j})$ $-48 \cdot 19 = -38\,\omega_{DE}$

$\omega_{DE} = 24\,s^{-1}\,\check{k}$

$$a_B = \overset{0}{a_A} + \overset{0}{\gamma_{AB}} \wedge (B-A) + \omega_{AB} \wedge [\overbrace{\omega_{AB} \wedge (B-A)}^{-380\,i}]$$

$$a_B = \begin{vmatrix} i & j & k \\ 0 & 0 & -19 \\ -380 & 0 & 0 \end{vmatrix} = 7220\,j$$

$$\rightarrow a_D = a_B + \underbrace{\gamma_{DB} \wedge (D-B)}_{\text{III}} + \underbrace{\omega_{DB} \wedge [\underbrace{\omega_{DB} \wedge (D-B)}_{380\,i - 912\,j}]}_{\text{IV}}$$

III
$$\begin{vmatrix} i & j & k \\ 0 & 0 & \gamma_{DB} \\ -48 & -20 & 0 \end{vmatrix} = 20\gamma_{DB}\,i - 48\gamma_{DB}\,j$$

IV
$$\begin{vmatrix} i & j & k \\ 0 & 0 & 19 \\ 380 & -912 & 0 \end{vmatrix} = 17328\,i + 7220\,j$$

$$a_D = 7220\,j + 20\gamma_{DB}\,i - 48\gamma_{DB}\,j + 17328\,i + 7220\,j$$

$$\underbrace{a_D = i\,(17328 + 20\gamma_{DB}) + j\,(14440 - 48\gamma_{DB})}_{\text{V}}$$

$$\rightarrow a_D = \overset{0}{a_E} + \underbrace{\gamma_{DE} \wedge (D-E)}_{\text{VI}} + \underbrace{\omega_{DE} \wedge [\underbrace{\omega_{DE} \wedge (D-E)}_{-912\,j}]}_{\text{VII}}$$

VI
$$\begin{vmatrix} i & j & k \\ 0 & 0 & \gamma_{DE} \\ -38 & 0 & 0 \end{vmatrix} = \boxed{-38\gamma_{DE}\,j}$$

VII $\begin{vmatrix} \check{i} & \check{j} & \check{k} \\ 0 & 0 & 24 \\ 0 & -912 & 0 \end{vmatrix} = 21888\,\check{i}$

$\boxed{a_D = 21888\,\check{i} - 38\,\gamma_{DE}\,\check{j}} \rightarrow$ VIII

V = VIII

$\check{i})\quad 17328 + 120\,\gamma_{DB} = 21888$

$$\boxed{\gamma_{DB} = 228\ s^{-2}\ \check{k}}$$

$\check{j})\quad 14440 - 48\,\gamma_{DB} = -38\,\gamma_{DE}$

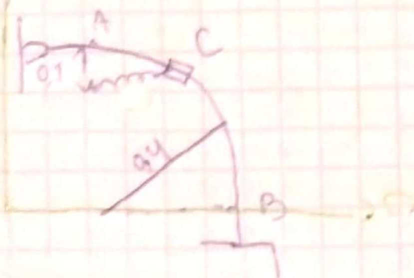
$$3496 = -38\,\gamma_{DE}$$

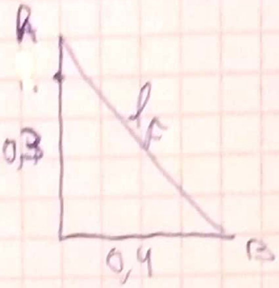
$$\boxed{\gamma_{DE} = -92\ s^{-2}\ \check{k}}$$

Colla

Collarin $m=4Kg$ $V_A=4 m/s$. Sin roz ¿Vel B? C

resorte = $k=400 N/m$ $l_0=0,2 m$





$l_F = \sqrt{0,3^2+0,4^2}$

$l_F = 0,5$

$\Delta x_F = 0,5-0,2 = 0,3$

→ No hay roz ni Fzas externas =

$$\Delta Em = 0$$

$$E_A = E_B$$

E_A → $V=4 m/s \Rightarrow E_c = \frac{1}{2} . 4 . 4^2 = 32 J$ $\Delta x_A = 0,1$ → comprimido

→ $E_{pg} = m.g.h = 4 . 9,8 . 0,4 = 15,68 J$

→ $E_e = \frac{1}{2} k \Delta x^2 = 200 . (-0,1)^2 = 2 J$

E_B → Sin h

→ $E_e = 200 . 0,3^2 = 18 J$

→ $E_c = \frac{1}{2} m v^2 = 2V^2$

$$32 + 15,68 + 2 = 18 + 2v^2 \Rightarrow \boxed{V = 3,98 m/s}$$ → RTA